

Answer Key & Teacher Notes

Session 1 – Equations from Tables • Session 2 – Equations from Situations

HOW TO USE THIS LESSON'S PAGES

one lesson = two sessions

Warm-Up	First 5–8 minutes of Session 1. Key: Unit Warm-Up Key.
Prerequisite Practice A/B	Before the lesson, for students who need the skill the session uses. Key in that folder.
Guided Notes	During the lesson. Students fill in the blanks as you teach; answers below.
Practice Form A	In class, after the lesson. Two pages. The self-check code tells students if they are on track.
Practice Form B	Same skills, new numbers: reteach, retake, small group or absent students.
Exit Ticket	Last 5 minutes. Two per page; cut in half.
Homework Packet	Sheet 1 after Session 1, Sheet 2 after Session 2. Level 1 has the same problems and answers, with a word bank, step frames and larger type for students who need language support.

SESSION 1 • TEACHER NOTES

Equations from Tables

TALK PROMPTS

- How does a table help you spot the constant rate in a relationship?
- How can you tell from the table alone that a rule multiplies instead of adds?

PARTNER TASK

Choir Risers. A table shows x : 1, 2, 3, 4 risers and y : 7, 14, 21, 28 singers. One partner writes the equation; the other tests it on two rows and uses it to find the singers on 11 risers (77).

WATCH FOR

- Writing a rule that fits only the first row ($y = x + 24$ for x : 3, 5, 10 and y : 27, 45, 90).
- Switching the variables ($x = 7y$ instead of $y = 7x$).
- Using the jump in y between rows as the rate when x does not go up by 1.

SESSION 2 • TEACHER NOTES

Equations from Situations

TALK PROMPTS

- How are the equation and the graph two views of the same relationship?
- What part of Zuri's equation would change if she raised her price to \$24 for each car?

PARTNER TASK

Hall and Bounce House. The hall charges a \$75 fee plus \$30 an hour; the bounce house charges a \$60 fee plus \$25 an hour. Each partner writes one equation and finds the cost for 4 hours (hall \$195, bounce house \$160), then together compare the fees and the hourly rates.

WATCH FOR

- Adding the rate instead of multiplying ($c = n + 2$ for buttons that cost \$2 each).
- Multiplying the fee by the variable ($c = 75h + 30$ instead of $c = 30h + 75$).
- Combining the fee and the rate into one rate ($c = 105h$).

ITEM ANALYSIS

depth of knowledge (DOK) for each practice item

Sheet	Standard	1	2	3	4	5	6	7	8	9	10	11
S1 • Form A	6.EE.C.9	1	2	1	1	2	3	2	3	3	1	1
S1 • Form B	6.EE.C.9	1	2	1	1	2	3	2	3	3	1	1
S2 • Form A	6.EE.C.9	1	2	1	2	1	3	2	2	3	1	1
S2 • Form B	6.EE.C.9	1	2	1	2	1	3	2	2	3	1	1

Items before the last three are Practice; then Find the Error (DOK 3) and two Test Practice items.

GUIDED NOTES

equation: A math sentence with an **equal** sign.

constant: A number that **never** changes.

rate: The amount for **each** one unit.

- If $y \div x$ is the **same** in every row, the equation is $y =$ (that number) $\times x$.
- If $y - x$ is the same in every row, the equation is $y = x +$ (that number).
- Check the equation with **two** rows.

Example: $6 \div 1 = 6$, $12 \div 2 = 6$, $30 \div 5 = 6$ The rate is **6** in every row. Answer: $y = 6x$. Check: $6 \times 5 = 30$.

YOU TRY

1. a. $y = 5x$ b. $y = x + 8$
2. $y = 8x$; $8 \times 12 = 96$

EXIT TICKET

1. a. $y = 4x$ b. $y = x + 5$
2. $y = 5x$; $5 \times 13 = 65$ cans
3. **B** (every row)

PRACTICE FORM A

1. a. $y = 5x$ b. $y = 12x$ c. $y = x + 3$ d. $y = 9x$
2. $y = 8x$; 7 passes cost $8 \times 7 = \$56$
3. $y = 5x$; $5 \times 12 = 60$ tiles
4. $y = 30x$; $x =$ gallons of gas
5. $y = 7x$; $7 \times 14 = \$98$
6. For $x = 5$: $5 + 24 = 29$, not 45, so it only fits the first row. Each y is 9 times x : $y = 9x$.
7. $y = 25x$; $x =$ number of packs, $y =$ number of beads
8. y goes up 4 each time, and $x = 0$ would give 5: $y = 4x + 5$. When $x = 10$, $y = 4 \times 10 + 5 = 45$.
9. He multiplied the wrong variable. Each y is 7 times x : $7 \times 2 = 14$. The correct equation is $y = 7x$.
10. **B** ($y = 5x$)
11. **C** ($y = 3x + 1$)

Self-check word **FACTOR**

PRACTICE FORM B

1. a. $y = 6x$ b. $y = 11x$ c. $y = x + 7$ d. $y = 4x$
2. $y = 9x$; 8 passes cost $9 \times 8 = \$72$
3. $y = 7x$; $7 \times 11 = 77$ singers
4. $y = 24x$; $x =$ cases of bottles
5. $y = 4x$; $4 \times 21 = \$84$
6. For $x = 4$: $4 + 14 = 18$, not 32, so it only fits the first row. Each y is 8 times x : $y = 8x$.
7. $y = 6x$; $x =$ number of stacks, $y =$ number of lockers
8. y goes up 3 each time, and $x = 0$ would give 4: $y = 3x + 4$. When $x = 10$, $y = 3 \times 10 + 4 = 34$.
9. He multiplied the wrong variable. Each y is 8 times x : $8 \times 3 = 24$. The correct equation is $y = 8x$.
10. **C** ($y = 12x$)
11. **A** ($y = 2x + 5$)

Self-check word **VALUES**

GUIDED NOTES

unit rate: The amount for exactly **1** unit.

multiplicative rule: **Multiply** the input by the same number.

additive rule: **Add** the same number to the input.

per: Means “for **each** one.”

- A rate (“each,” “per”) **multiplies** the variable.
- A one-time fee or a starting amount is **added** once.

Example: Define the letters: d = days, e = **dollars earned**. The rate is \$9 for each **day**, so $e = 9d$. Answer: Test $d = 5$: $e = 9 \times 5 = 45$ dollars.

YOU TRY

1. $a. p = 8b$ $b. c = 20m + 10$
2. $p = 13n$; $13 \times 6 = \$78$

EXIT TICKET

1. $a. c = 6t$ $b. k = j + 4$
2. $c = 35n$; $35 \times 9 = \$315$
3. **B** ($c = 10h + 25$)

PRACTICE FORM A

1. $a. c = 11h$ $b. c = 14n$ $c. m = b + 7$ $d. c = 5h + 20$
2. $a.$ independent: hours, h ; dependent: cost, c $b. c = 9h$ $c. 9 \times 6 = \$54$
3. $p = 13n$; $13 \times 8 = \$104$
4. $m = b + 7$; $15 + 7 = 22$ years old
5. $r = 48b$; $48 \times 6 = 288$ riders
6. $A: y = 5x$ (multiply by 5). $B: y = x + 4$ (add 4). A multiplies x by a rate; B adds the same amount to x .
7. $a. s = 45p$ $b. s = 45p + 60$ $c. 45 \times 4 + 60 = \$240$
8. $c = 30h + 75$; $30 \times 5 + 75 = \$225$
9. Her rule adds 2 once. Test 5 buttons: $5 + 2 = 7$, but 5 buttons cost $2 \times 5 = \$10$. The price repeats for each button, so multiply: $c = 2n$.
10. **B** ($c = 9h$)
11. **C** ($c = 5h + 20$)

Self-check word **SYMBOL**

HOMEWORK PACKET • ANSWERS

same answers for the core packet and Level 1

SHEET 1 • EQUATIONS FROM TABLES

1. $a. y = 7x$ $b. y = x + 9$ $c. y = 5x$ $d. y = x - 3$ $e. y = x + 6$ $f. y = 9x$ $g. y = 8x$ $h. y = x - 5$
2. $y = 11x$; 66, 99
3. Each y is 2 less than x : $y = x - 2$. When $x = 20$, $y = 18$.
4. $y = 4x$; 10 tickets cost \$40
5. It only works for the first row: $3 \times 4 = 12$, not 8. Each y is 4 more than x , so $y = x + 4$.

PRACTICE FORM B

1. $a. c = 12h$ $b. c = 15t$ $c. l = s - 4$ $d. c = 8h + 30$
2. $a.$ independent: hours, h ; dependent: cost, c $b. c = 11h$ $c. 11 \times 7 = \$77$
3. $e = 21n$; $21 \times 7 = \$147$
4. $m = c + 5$; $12 + 5 = 17$ years old
5. $r = 14v$; $14 \times 8 = 112$ riders
6. $A: y = 3x$ (multiply by 3). $B: y = x + 2$ (add 2). A multiplies x by a rate; B adds the same amount to x .
7. $a. s = 35p$ $b. s = 35p + 50$ $c. 35 \times 6 + 50 = \$260$
8. $c = 25h + 60$; $25 \times 4 + 60 = \$160$
9. Her rule adds 3 once. Test 4 stickers: $4 + 3 = 7$, but 4 stickers cost $3 \times 4 = \$12$. The price repeats for each sticker, so multiply: $c = 3n$.
10. **C** ($c = 14n$)
11. **A** ($c = 6h + 15$)

Self-check word **SOLVED**

SHEET 2 • EQUATIONS FROM SITUATIONS

1. $a. y = 12x$ $b. y = x + 3$ $c. y = 15x + 20$
2. $y = 5x + 4$; 4 games: $5 \times 4 + 4 = \$24$
3. $y = 2x + 3$; 7 miles: $2 \times 7 + 3 = \$17$
4. \$30, \$38, \$46, \$70; $y = 8x + 30$
5. She added the fee to the rate, so every hour costs \$35. The fee is paid once: $y = 10x + 25$.
6. **B** ($c = 25h + 40$)
7. **C** (Start with 12 gal, add 3 gal per min)